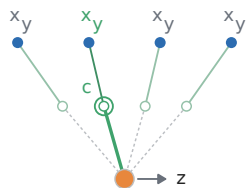
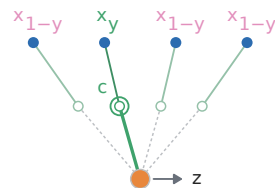


A

Context-gated branch-conflict task

compatible: $\chi = 0$ 

same update sign

conflicting: $\chi = 1$ 

opposing update signs

B

Forward selection, backward credit

local eligibility

 $e_b \neq 0$ for every branch

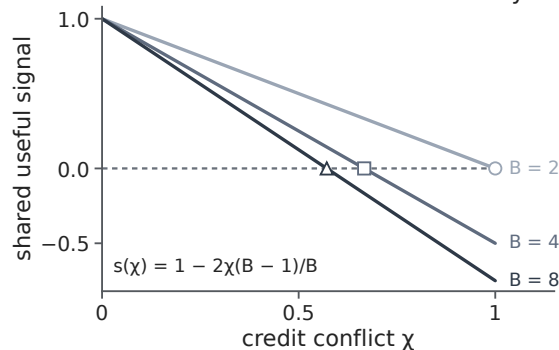
branch-specific credit

 $\delta_b = \delta \cdot 1[b = c]$ 

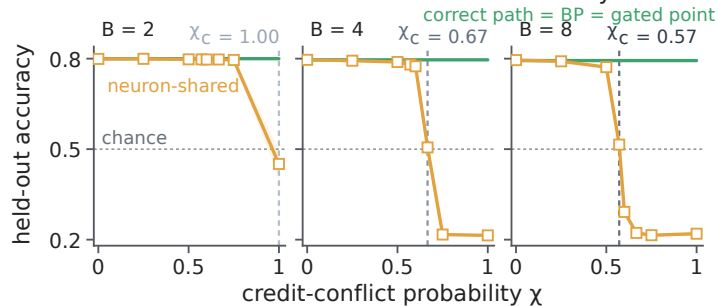
neuron-shared credit

 $\delta_b = \delta / B$ for every branch**C**

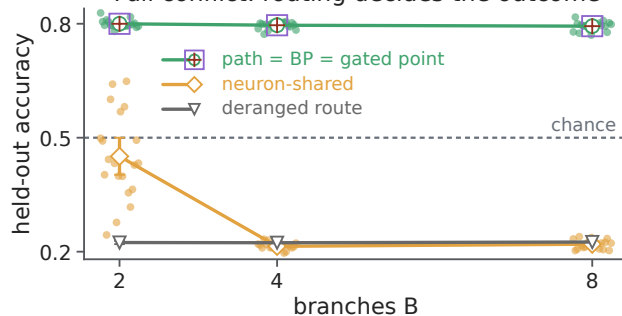
Predicted shared-mode boundary

**D**

Trained transition follows the boundary

**E**

Full conflict: routing decides the outcome

**F**

Observed collapse follows the prediction

